

Probability and Statistics using R

Lab Session-2

Data Objects, Vectorization, Functions and Data Frames

Course Code : MAT445

Date:23.07.2026

Time : 3:15PM - 4:15 PM

Programming Tasks

1. Create numeric, character, and logical variables. Display their values and data types.
2. Create a list containing a student's Name, Age, CGPA, and Department. Access each element separately.
3. Create two vectors of 10 numbers each. Perform element-wise addition, subtraction, multiplication, and division.
4. Given a vector of marks, add 5 grace marks to every student using vectorized operations and display the updated marks.
5. Create a vector of numbers from 1 to 20. Display only the even numbers using vectorized logical indexing.
6. Write a user-defined function to calculate the area of a circle.
7. Write a function that accepts a vector and returns its maximum, minimum, and average values.
8. Write a recursive function to calculate the factorial of a given number.
9. Create a data frame containing Student ID, Name, Marks, and Department. Display students scoring more than 80 marks.
10. Create a data frame of employees with Name, Salary, and Experience. Add a new column named Bonus equal to 10% of the salary.

Answers

1. Data Objects

```
1 num <- 25
2 name <- "R Programming"
3 flag <- TRUE
4
5 print(num)
6 print(name)
7 print(flag)
8
9 class(num)
10 class(name)
11 class(flag)
```

2. Data Objects

```
1 student <- list(
2   Name = "Arun",
3   Age = 20,
4   CGPA = 8.9,
5   Department = "CSE"
6 )
7
8 print(student$Name)
9 print(student$Age)
10 print(student$CGPA)
11 print(student$Department)
```

3. Vectorization

```
1 v1 <- c(2,4,6,8,10,12,14,16,18,20)
2 v2 <- c(1,2,3,4,5,6,7,8,9,10)
3
4 v1 + v2
5 v1 - v2
6 v1 * v2
7 v1 / v2
```

4. Vectorization

```
1 marks <- c(65, 78, 84, 91, 56, 73)
2
3 updated_marks <- marks + 5
4
5 print(updated_marks)
```

5. Vectorization

```
1 numbers <- 1:20
2
3 even_numbers <- numbers[numbers %% 2 == 0]
4
5 print(even_numbers)
```

6. Functions

```
1 area_circle <- function(r)
2 {
3   area <- pi * r^2
4   return(area)
5 }
6
7 area_circle(7)
```

7. Functions

```
1 statistics <- function(x)
2 {
3   cat("Maximum =", max(x), "\n")
4   cat("Minimum =", min(x), "\n")
5   cat("Average =", mean(x), "\n")
6 }
7
8 marks <- c(75, 82, 91, 68, 88)
9
10 statistics(marks)
```

8. Functions

```
1 factorial_fun <- function(n)
2 {
3   if(n == 0)
4     return(1)
5   else
6     return(n * factorial_fun(n - 1))
7 }
8
9 factorial_fun(5)
```

9. Data Frames

```
1 students <- data.frame(  
2   ID = c(101,102,103,104),  
3   Name = c("Arun","Bala","Divya","Kiran"),  
4   Marks = c(78,91,84,69),  
5   Department = c("CSE","ECE","IT","MECH")  
6 )  
7  
8 high_marks <- students[students$Marks > 80, ]  
9  
10 print(high_marks)
```

10. Data Frames

```
1 employees <- data.frame(  
2   Name = c("Amit","Ravi","John","Sara"),  
3   Salary = c(35000,42000,50000,46000),  
4   Experience = c(2,5,7,4)  
5 )  
6  
7 employees$Bonus <- employees$Salary * 0.10  
8  
9 print(employees)
```